

EMERGENT SPIRAL ARM PITCH ANGLES IN THE PHASE INTERFERENCE UNIVERSE MODEL (PIUM)

Phase Geometry, Minimal Dynamics, and an Observational Linkage Framework

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ABSTRACT

Within the Phase Interference Universe Model (PIUM), spiral galaxy arms are interpreted as phase-coherent interference structures emerging from a universal wave field Φ . This work develops a structured framework for spiral-arm pitch angles (ψ) by separating three layers: analytic phase geometry, minimal phenomenological dynamics, and a morphology–dynamics linkage protocol.

The geometric layer shows that logarithmic spiral structure follows directly from a minimal phase ansatz via $\tan \psi = m/k$. The dynamical layer explores whether stable spiral patterns and mild radial pitch-angle variation can arise under a simple coherence-driven evolution; results are presented as illustrative numerical experiments.

Because the SPARC database supplies a robust kinematic baseline (outer rotation-curve slopes) but does not contain pitch-angle measurements, this study defines a transparent protocol for cross-matching external pitch-angle catalogues with the accepted SPARC sample and outlines a reproducible procedure for future direct measurements on imaging data. No claim of observational validation is made at this stage. The framework provides a clear, auditable pathway for systematic tests between pitch angle and SPARC kinematic parameters (V_{flat} , outer slope s).

1. INTRODUCTION

The pitch angle ψ is a primary quantitative descriptor of spiral structure in disk galaxies, with observed values typically spanning $\approx 5^\circ$ – 40° . While empirical correlations with morphology and mass are well documented (Yu & Ho 2019, 2020; Reshetnikov et al. 2023), a unified physical origin remains an open question.

In the PIUM framework, spiral arms are understood as large-scale interference maxima within a universal phase field Φ . Consequently, the pitch angle is expected to emerge from the phase topology itself.

This paper constructs a coherent and reproducible framework linking (i) analytic phase geometry, (ii) minimal phenomenological dynamics, and (iii) an observational linkage protocol anchored to the existing SPARC kinematic baseline. The emphasis is on transparency and falsifiability rather than claiming empirical confirmation.

2. PHASE-GEOMETRIC ORIGIN

The universal field is expressed as

$$\Phi(r, \theta, t) = A(r, t) \exp(i \phi(r, \theta, t)).$$

A minimal logarithmic phase structure for an m -armed pattern is

$$\phi(r, \theta) = m \theta - k \ln(r / r_0).$$

Constant-phase loci yield the logarithmic spiral equation

$$\theta = (k/m) \ln r + C,$$

with the pitch angle

$$\tan \psi = m / k \quad \Rightarrow \quad \psi = \arctan(m / k).$$

For $m = 2$, a broad phenomenological range of k produces pitch angles consistent with the observed distribution in spiral galaxies. This relation is purely geometric.

3. MINIMAL PHENOMENOLOGICAL DYNAMICS

Temporal behaviour is explored using the minimal evolution equation

$$i \partial \Phi / \partial t \approx -\alpha \nabla^2 \Phi - i \gamma (\phi - \langle \phi \rangle) \Phi,$$

where the second term is a phenomenological coherence constraint. Numerical simulations starting from centrally concentrated initial conditions with weak perturbations show the formation of stable two-armed spiral patterns that persist over long evolution times and exhibit mild outward pitch-angle variation (of order a few degrees). These findings are illustrative demonstrations within a minimal model.

4. SPARC KINEMATIC BASELINE

The kinematic reference is the re-analysed SPARC sample of 79 galaxies satisfying $Q \leq 2$, with the outer region defined as $R > 2.5 R_{\text{disk}}$ and $N_{\text{outer}} \geq 8$. The primary quantity is the logarithmic outer slope

$$s = d \log_{10}(V_{\text{obs}}) / d \log_{10}(R),$$

with median value near zero, consistent with predominantly flat outer rotation curves. This baseline supplies V_{flat} , R_{disk} and outer slope, but contains no pitch-angle information.

5. MORPHOLOGY–DYNAMICS LINKAGE PROTOCOL

To connect morphology and kinematics, a two-stage protocol is defined. First, published pitch-angle measurements (primarily from Yu & Ho 2019) are cross-matched to the accepted SPARC sample after standardized name normalization. Second, a direct measurement pipeline is specified for future application to SPARC imaging data, including fixed deprojection, polar coordinate transformation, ridge tracing, and sliding-window logarithmic fitting. All steps are designed to be deterministic and uniformly applied, with full documentation of selection criteria and quality cuts.

6. INTERPRETATION AND OUTLOOK

The framework establishes a geometric basis for logarithmic spirals, demonstrates dynamical plausibility of stable patterns under minimal coherence constraints, and provides a clear observational linkage path. Limitations include image-quality dependence and the phenomenological character of the current dynamical realization.

Future work will focus on constructing the matched dataset, executing the direct measurement pipeline, and performing statistical morphology–dynamics tests.

7. CONCLUSION

This work defines a transparent and reproducible framework connecting PIUM phase geometry, minimal dynamics, and observational galaxy data. By clearly separating the analytic, numerical, and linkage layers, it offers a structured foundation for future empirical testing of the model against spiral-arm morphology and kinematics.

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